DO NOW — RECALL

Lesson 116 · Linkers, Loaders and Program Libraries

Name: _____ Class: _____ Date: _____

1. Which compilation stage converts source characters into tokens?

2. In a BST, where is a value less than the current node placed?

3. Why can an array provide direct indexed access?

4. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Lexical analysis.

2. Model answer

In its left subtree (under the chosen no-duplicate convention).

3. Model answer

Its element locations can be calculated from base address and index.

4. Model answer

Sum 0 and carry-out 1.

By the end, you can…

• Source files become object modules with symbols/references.

• Linker resolves symbols and combines modules/libraries; static/dynamic linkage differs.

• Loader maps executable and needed libraries into memory, then begins execution.

Vocabulary — know these terms

linker — Resolves symbols and combines object modules.

loader — Places an executable in memory ready to run.

library — Reusable compiled routines/functions.

dynamic linking — Resolution/use of shared code at load time or runtime.

NEW LEARNING

Read, annotate and explain

Compile modules

Source files become object modules with symbols/references.

Link dependencies

Linker resolves symbols and combines modules/libraries; static/dynamic linkage differs.

Load and run

Loader maps executable and needed libraries into memory, then begins execution.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

main.c calls a function in maths.c. Explain the steps from separate source files to a running program.

Model answer

Compile source modules into object files; linker resolves the call and produces linked executable (or dynamic references); loader places executable and required libraries in memory and starts execution.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the roles of linker and loader with examples. [6]

Question 2

Distinguish static from dynamic linking, giving one trade-off of each. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A deployed app fails because a required shared library is unavailable. Explain why compile success did not prevent this. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A successful compilation guarantees that every dynamically linked library will be available at runtime.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.